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Withdraw Money and Receipt in ATM

1. AIM

To design and study the UML Sequence Diagram and UML Class Diagram for an ATM Withdrawal System that withdraws an amount and generates a receipt.

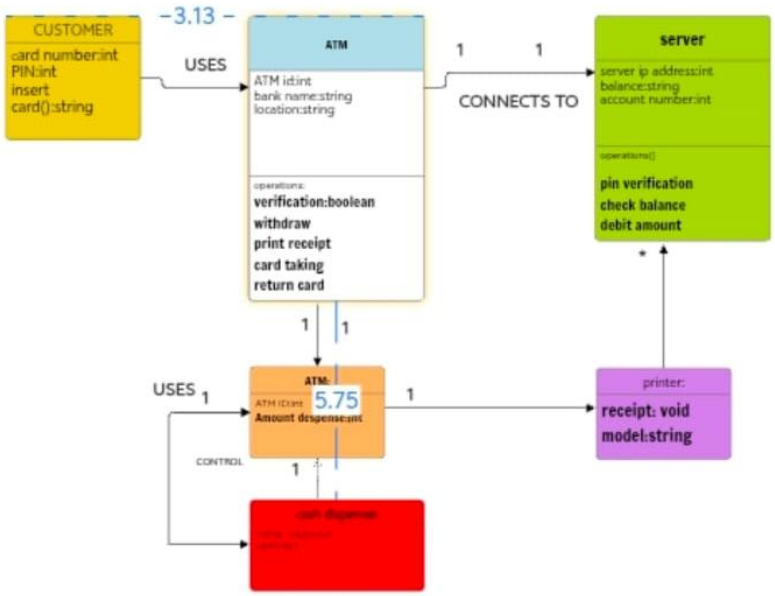
2. INTRODUCTION

An ATM (Automated Teller Machine) allows users to perform banking operations such as withdrawing money and printing receipts.

In this experiment:

- The **Sequence Diagram** explains how the system works step-by-step.
- The **Class Diagram** explains the structure of the system and relationships between classes.

3. UML CLASS DIAGRAM



Explanation of Class Diagram

The Class Diagram shows the static structure of the ATM system.

It includes the following classes:

- User
- ATM UI
- Withdrawal
- Bank Server
- Account
- ATM Machine

Each class contains attributes (data members) and methods (functions).

Working of Class Diagram

The working of the class diagram is explained below:

1. User Class

The User class contains user details like name and cardID. It performs actions such as:

- insertcard()
- enterPIN()
- withdrawal()

The user interacts with the ATM UI to initiate transactions.

2. ATM UI Class

The ATM UI class acts as an interface between user and system. It:

- Displays menu options
- Prompts for amount
- Shows error messages

It communicates with Bank Server and Withdrawal classes.

3. Withdrawal Class

The Withdrawal class handles the withdrawal transaction. It:

- Validates the balance
- Executes the transaction

It ensures sufficient balance before completing the process.

4. Bank Server Class

The Bank Server:

- Authenticates user details
- Authorizes transactions

It confirms whether the transaction is allowed or not.

5. Account Class

The Account class stores:

- Account number
- Balance

It provides:

- debit(amount)
- getbalance()

It updates the account balance after successful withdrawal.

6. ATM Machine Class

The ATM Machine:

- Dispenses cash
- Prints receipt

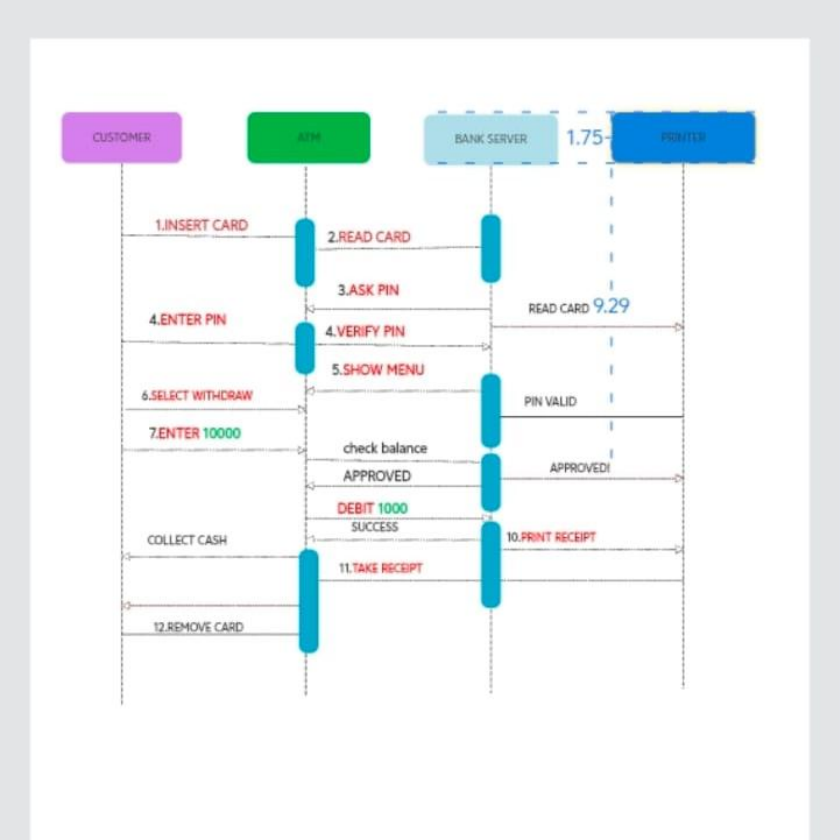
It represents the physical ATM hardware.

Overall Working of Class Diagram

1. User interacts with ATM UI.
2. ATM UI communicates with Bank Server for authentication.
3. Withdrawal class checks account balance.
4. Bank Server authorizes transaction.
5. Account balance is debited.
6. ATM Machine dispenses cash and prints receipt.

Thus, the Class Diagram represents the structural organization of the ATM system.

4. UML SEQUENCE DIAGRAM



Explanation of Sequence Diagram

The Sequence Diagram shows the interaction between:

- User
- ATM UI
- Bank Server

It explains how messages flow between them during a withdrawal transaction.

Working of Sequence Diagram

The working of the sequence diagram is as follows:

1. The User inserts the ATM card.
2. The ATM UI sends card details to the Bank Server for validation.
3. The Bank Server verifies the card and sends a valid response.
4. The ATM displays the home screen.
5. The User selects the withdraw option.
6. The ATM asks the user to choose the account type.
7. The User selects the savings account.
8. The ATM prompts the user to enter the withdrawal amount.
9. The User enters the amount (example: 10000).
10. The ATM UI sends a balance check request to the Bank Server.
11. The Bank Server verifies the balance and sends approval.
12. The ATM dispenses cash and prints the receipt.

Thus, the Sequence Diagram represents the dynamic behavior of the ATM system.

5. RESULT

Thus, the UML Sequence Diagram and UML Class Diagram for the ATM Withdrawal System were successfully designed and their working was studied separately.